Motorola M6800 Support Products

For Total Microcomputer System Design

The evolution of large-scale integration (LSI) has spawned the microprocessor as a low-cost circuit component, and has created a viable alternative to the conventional hardwired method of circuit, and system design. The resultant concept of utilizing a programmable microcomputer to perform control functions previously relegated to dedicated, single-function circuitry presents such a powerful approach to the implementation of control systems that it is widely acclaimed as the single most important development of the past decade.

To facilitate the design of pervasive and cost-effective microcomputer-based systems, Motorola has generated a repertoire of basic LSI building blocks. Appropriate combination of these components permits the establishment of an end-system with any desired degree of sophistication. Known as the M6800 Microcomputer Family, this repertoire at present includes the following components:

MC6800 – A Microprocessing Unit (MPU) which represents the heart of a microcomputer system. Encompassed within this component are the arithmetic and control capabilities that are fundamental to any computer system.

MC6820 – A Peripheral Interface Adapter (PIA) that interfaces the MPU with a variety of input/output peripherals such as terminals, printers, etc. Its parallel I/O architecture interfaces with a parallel oriented peripheral.

MC6850 – An Asynchronous Communications Interface Adapter (ACIA) for communications between the MPU and a serially oriented peripheral, such as a Modem. This converts serially transmitted data to the parallel orientation required by the MPU, and vice versa.

MC6852 – A Synchronous Serial Data Adapter (SSDA) for communications between the MPU and a bit-serial peripheral.

MC6860 – A low-speed Modem (modulator-demodulator) that permits the MPU to be accessed, through the ACIA, by data transmitted over the telephone lines.

MCM6830 – A 1024 x 8-bit Read-Only Memory (ROM), programmed by the factory with the appropriate instructions stipulated by the designer's end-system requirements.

MCM6810 – A 128 x 8-bit Random Access Memory (RAM) with access time as low as 350 ns (MC6810AL1).

All of the above parts are mutually compatible in a bus-oriented system that requires no additional buffers or other outboard components. An advanced N-channel MOS technology yields reliable operation while a powerful 72-statement instruction set provides a high degree of programming flexibility.

But successful microcomputer-based system design requires more than a set of compatible "hardware" components. Even after a "working computer" has been implemented through the interconnection of the required components, there still remains the development and verification of the "program" that converts a mindless computer into a dedicated functional system. The M6800 system, therefore, comprises an extensive and growing array of support products dedicated to insuring the timely and successful completion of a design effort.

The Support Products

Motorola's total M6800 System Design Support concept encompasses the following categories:

Support Hardware – A series of instruments that emulate, demonstrate and evaluate the operation of any M6800-based design.

Support Peripherals – A growing array of peripheral equipment that interfaces with M6800 system designs to increase data throughput during system development.

Support Software – Development and diagnostic programs that permit the design of end-use programs utilizing Time-Share computers, in-house computers or the support hardware mentioned above.

Design Support – Includes documentation, training, applications assistance and a staff of professionals to answer questions and help solve system development problems.

The information to follow describes, in detail, these support products.

MOTOROLA MICROSYSTEMS

ISSUE A

Ⓜ **MOTOROLA** *MICROSYSTEMS*

The use of microprocessors can greatly simplify many hardware designs. However, a microprocessor-based design includes software, an element not found in conventional hard-wired designs. The proper tools are essential to the design of a microprocessor system.

Although microprocessor software can be written directly in machine language, this is too tedious, time-consuming, error-prone, and expensive for all but the smallest system.

Most microprocessor software is presently written in assembly language. At this level, the programmer has complete and explicit control of the microprocessor, while the assembler relieves him of many burdensome details. High-level languages (compilers) are becoming increasingly popular for programming microprocessors. Although such programs are typically somewhat larger and slower than those written in assembly language, they can be developed more rapidly and maintained more easily.

Motorola provides software development tools for three environments:

1. Resident Software – Use of the EXORciser for software development permits dual use of this system development equipment. This low-cost alternative for software development employs the same prototyping hardware used for program test and hardware/software integration. Software development tools also are available for Motorola's Evaluation Modules.
2. In-House Computer Systems – Cross-computer software provides low-cost program development and simulation facilities for in-house computer systems ranging from minicomputers to large mainframe machines. Capabilities of an in-house system may include powerful system software, high-speed peripherals, and multiple-user access.
3. Time-Sharing – The user has immediate worldwide access to M6800 software development packages. Time-sharing provides access to a powerful user-oriented system at a minimal initial cost. It may also be used in meeting workload peaks and critical schedules.

M6800 Support Software

MOTOROLA MICROSYSTEMS

Motorola provides total design support to customers for the development of M6800-based microcomputer systems. This support includes:

- Documentation
- Training
- Application Assistance

M6800 Design Support

Design Support

MOTOROLA Semiconductor Products Inc.

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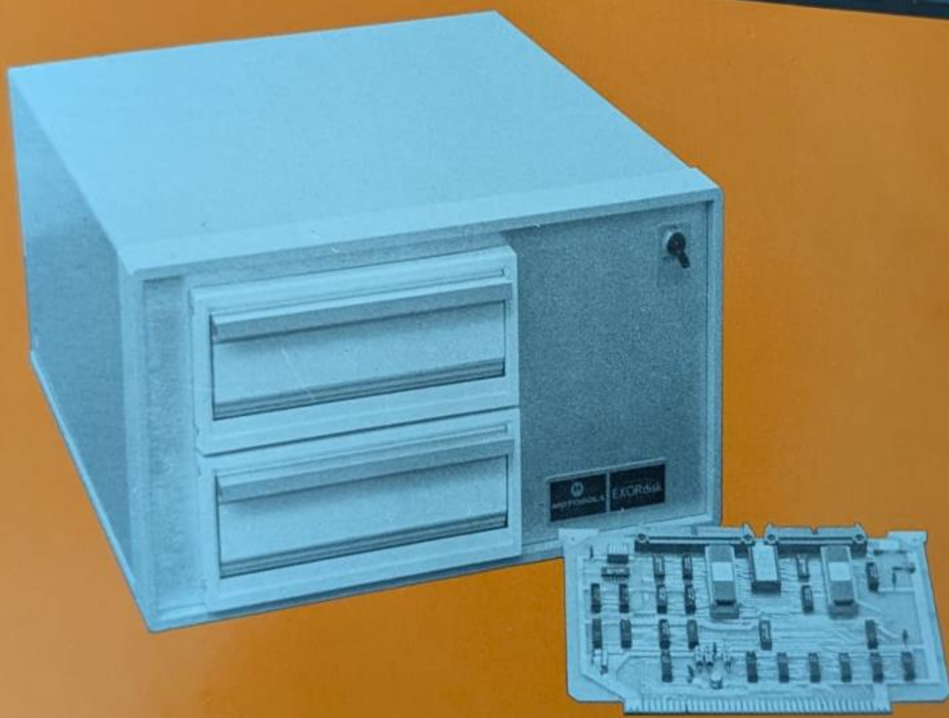
MOTOROLA MICROSYSTEMS

The only peripheral required to use the EXORciser for design purposes is a data terminal. For major design efforts, however, considerable design time can be saved with additional peripherals that speed up program development. EXORdisk and EXORtape are among these development tools.



M6800 Support Peripherals

MOTOROLA *MICROSYSTEMS*

Motorola has introduced a series of products to aid in the development of microprocessor systems. These products not only simplify breadboarding of an M6800-based microcomputer system, but also provide an efficient and economical means of evaluating, monitoring, and troubleshooting the system and evaluating and testing the M6800 family of LSI parts being used in the system.

M6800

Support Hardware

M6800 Microcomputer System Support Products

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*Data Sheet Available Soon.